



M-Tech Production & Marketing

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Experiment 4

Lab Task 2

Internship Program — Batch 2026

Submitted By:	Abdullah Rafique (MTech-AI260120)
Department:	AI / ML
Submitted To:	Ceo Moazzan
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1. Experiment 4 – Lab Tasks

File: *Experiment_4.py*

2. Viva Questions

1. How is Machine Learning different from Traditional Programming?

Machine Learning differs from traditional programming in the way problems are solved. In traditional programming, a programmer writes explicit rules and instructions that the computer follows to produce the desired output. In contrast, Machine Learning enables the computer to learn patterns and relationships directly from data without being explicitly programmed with every rule.

In traditional programming:

- **Input + Rules → Output**

In Machine Learning:

- **Input + Correct Output (Training Data) → Machine Learning Model**

The trained model then learns the underlying patterns and can make predictions on new, unseen data.

2. What are the main types of Machine Learning?

Machine Learning is broadly classified into three main categories:

Supervised Learning

Supervised learning uses **labeled data**, where both the input features and the correct output labels are known. The model learns the relationship between inputs and outputs to make predictions.

Examples:

- House price prediction (Regression)
- Email spam detection (Classification)

Unsupervised Learning

Unsupervised learning works with **unlabeled data** and aims to discover hidden patterns, relationships, or groups within the dataset.

Example:

- Customer segmentation using clustering.

Reinforcement Learning

Reinforcement learning enables an agent to learn through **trial and error** by interacting with an environment. The agent receives rewards for correct actions and penalties for incorrect ones, gradually learning the optimal strategy.

Examples:

- Game-playing AI
- Self-driving vehicles
- Robotics

3. What is the difference between a Feature and a Label?

A **feature** is an input variable that provides information used to make predictions, while a **label** is the target output that the model is trained to predict.

For example, in a **house price prediction** problem:

Feature	Label
Area of the house	House Price
Number of bedrooms	
Location	

During training, the Machine Learning model learns the relationship between the features and the corresponding label.

4. Why is data divided into Training and Test sets?

The dataset is divided into **training** and **test** sets to evaluate how well a Machine Learning model performs on unseen data.

- The **training set** is used to teach the model by allowing it to learn patterns from the data.
- The **test set** is used to evaluate the model after training.

If the model is evaluated using the same data it was trained on, it may simply memorize the data instead of learning meaningful patterns. Testing on unseen data provides a realistic measure of the model's ability to generalize to real-world situations.

5. Explain the basic Machine Learning workflow.

A standard Machine Learning workflow consists of the following steps:

1. Collect relevant data.
2. Clean and preprocess the data.
3. Split the dataset into training and testing sets.
4. Train the Machine Learning model.
5. Evaluate the model using appropriate performance metrics.
6. Tune hyperparameters to improve model accuracy.
7. Deploy the trained model for practical use.

Following this workflow helps ensure that Machine Learning models are accurate, reliable, and capable of making meaningful predictions.

6. Differentiate between Classification and Regression.

Classification and Regression are two major types of supervised learning problems.

Classification	Regression
Predicts categories or classes.	Predicts continuous numerical values.
Output is discrete.	Output is continuous.
Example: Spam or Not Spam	Example: House Price Prediction
Example: Disease Detection	Example: Temperature Prediction

The key difference is that **classification predicts categorical outputs**, whereas **regression predicts numerical values**.

3. Multiple Choice Questions (MCQs)

Q1. Predicting the price of a house is an example of:

Answer: (B) Regression

Q2. Labeled data is used in:

Answer: (A) Supervised Learning

Q3. The output that a Machine Learning model predicts is called:

Answer: (B) Label

Q4. A Machine Learning model is evaluated using the:

Answer: (B) Test Set

Q5. Grouping customers without labeled data is an example of:
Answer: (C) Clustering

4. Intern Information

Field	Details
Intern Name	Abdullah Rafique
Supervisor Name	Mr. Mohazin Zahid
Date	07 July, 2026
Domain	AI/ML Intern
Intern Signature	<u>ABD</u>